Cognitive Robustness and Privacy-Constrained Local Academic Assistance Under Domain Shift

Abstract

We study cognitive robustness and privacy-constrained deployment for a fully offline academic-assistance system under educational-domain shift. The implemented framework combines multimodal text/PDF ingestion, dense retrieval over FAISS, a Label Distribution Learning (LDL) Bloom classifier, uncertainty-aware Bloom gating, role-aware privacy screening, and local Qwen GGUF generation. Rather than treating these as independent features, the evaluation asks whether cognitive labels, ordinal structure, retrieval utility, and protected exam access policies remain reliable when moved across datasets and deployment roles.

Across Figshare and MoocRadar Bloom-labelled questions, directional transfer shows a sharp drop in exact reduced-label classification, especially in the ternary setting, but much stronger preservation of ordinal proximity. This motivates soft LDL feedback and human-in-the-loop fallback for uncertain or severe-jump-risk predictions. Privacy results are reported with the same caution: the InfoNCE retrieval-leakage proxy is a negative result, while the role-aware guard shows high measured resistance only under the defined adversarial prompt taxonomy. The contribution is therefore an evidence-backed analysis of domain-shift failure modes and local deployment constraints, not a claim of universal privacy or universal Bloom generalization.

Keywords — retrieval-augmented generation; educational NLP; domain shift; label distribution learning; Bloom's taxonomy; privacy-constrained deployment; ordinal evaluation; uncertainty-aware reasoning.

1. Introduction

University-scale AI assistants increasingly need to operate without sending student or institutional content to external services. Retrieval-augmented generation is a natural fit because it grounds compact local generators in retrievable context. We design and evaluate such a pipeline under four explicit constraints: (i) CPU-only inference, (ii) ≤ 1 GB private working-set memory, (iii) deterministic and reproducible execution, and (iv) no external API calls. The evaluation is intentionally austere — we report what the system actually does on the implemented benchmark, including a clearly negative result on the current privacy term. The paper is organised around two findings: (i) ordinal cognitive structure degrades more gracefully than exact labels under domain shift, and (ii) privacy controls should be evaluated as measurable safety-utility tradeoffs rather than binary guarantees.

This work does not introduce a new base model; instead, it demonstrates that ordinal cognitive structure is more stable than categorical labels under domain shift, and that privacy constraints in offline RAG induce measurable safety-utility tradeoffs.

2. Contributions

We make three evidence-oriented contributions framed as questions answered (not components assembled), around cognitive robustness and privacy-constrained deployment:

- **Directional cognitive-robustness evaluation.** We evaluate Bloom classification under Figshare ↔ MoocRadar transfer, explicitly reporting the 6-class-to-reduced-space mapping, exact-label degradation, within-one-level accuracy, and severe ordinal error rate (distance ≥ 2).
- **Uncertainty-aware deployment policy.** The LDL classifier is used as a distribution, not only as an argmax label: adjacent-level ambiguity is exposed as soft feedback, while low-confidence or severe-jump-risk cases are routed to generalized or human-in-the-loop handling, rather than forcing brittle Bloom-level specialization.
- **Finding + framework, not module stacking.** The contribution is a measurement framework that separates categorical degradation from ordinal degradation and couples bounded empirical privacy resistance with explicit

utility-cost reporting (safety-utility tradeoffs, not perfect guarantees).

• **Privacy evaluation with bounded claims.** We separate a negative retrieval-leakage sweep from a role-aware adversarial prompt taxonomy, and state privacy claims only as measured resistance under the defined attack set (including adaptive/multi-turn probes), not as perfect or general security.

3. Related Work

Recent work on retrieval-augmented generation (RAG) has highlighted both its effectiveness and its privacy risks. Studies on privacy issues in RAG show that retrieval can leak protected content through the selected passages themselves (Good & Bad RAG) and that privacy-preserving retrieval can be targeted with formal distance-based guarantees (RemoteRAG, DistanceDP). However, those approaches generally assume cloud-based or guarantee-first settings and do not directly address the offline, role-constrained university assistant deployment regime where privacy evidence is measured under a defined threat model rather than proven end-to-end.

Runtime mitigations (e.g., dynamic retrieval filtering and anonymization and perturbation-based privacy-preserving RAG attempt to reduce leakage at retrieval time, while multimodal retrieval can extend leakage pathways beyond text (Beyond Text). Our paper takes a complementary, threat-model-specific empirical approach: we combine a role-aware privacy guard with an explicit adversarial prompt taxonomy and report safety-utility tradeoffs under measurable attacks, avoiding over-strong claims of cryptographic or differential-privacy security.

On the educational side, Bloom/OBE classification is typically treated as flat classification and optimized for exact label accuracy. Yet domain shift can move topic vocabulary while preserving cognitive structure, motivating ordinal analyses. Our central novelty is to quantify ordinal robustness of Bloom classification under Figshare ↔ MoocRadar transfer and to couple this ordinal signal with uncertainty-aware deployment for an offline assistant. Feasibility at CPU scale is supported by efficient tiny-model and quantization research (QLoRA, AWQ, TinyLlama, Phi-2) and llama.cpp, but those works do not study Bloom-ordinal robustness under domain shift nor the associated privacy constraints.

4. System Architecture

The system has five modules (mapped 1-to-1 to source files):

• **Ingestion** (*ingestion.py*) — PyMuPDF text extraction, EasyOCR (lazy-loaded) for scanned/image input, plain-text loading, and deterministic token-window chunking.

• **Retriever** (*retriever.py*) — sentence-encoder embedding with all-MiniLM-L6-v2 (frozen, L2-normalised, dim 384), exact ANN over FAISS IndexFlatL2, and an InfoNCE-based privacy-aware re-ranking term.

• **Bloom-LDL Classifier** (*classifier.py*) — a linear LDL head on top of frozen MiniLM features, trained with Gaussian-smoothed soft labels combined with an ordinal-PMF target, an ordinal pairwise margin penalty, and an entropy regulariser.

• **Generator** (*models.py*) — Qwen-1.5B-Instruct (Q4_K_M GGUF) via llama-cpp-python, ChatML prompt with [BOUNDED CONTEXT] / [QUESTION] / [COGNITIVE LEVEL] / [INSTRUCTION] blocks, greedy decoding with temperature = 0, top_k = 1, top_p = 1, seed = 42, and an explicit llm.reset() before every call.

• **Uncertainty** (*uncertainty.py*) — Bloom-level normalised entropy $H(p)/\log K$ and Semantic Predictive Uncertainty (SPU) computed via chunk-subset perturbation, plus a confidence threshold gate that abstains or falls back when entropy, top-1 probability, or severe ordinal-jump mass indicates unreliable Bloom specialisation.

• **Privacy guard** (*privacy_guard.py*) — role-aware access control and output screening for protected exam uploads, evaluated with direct reconstruction, indirect leakage, paraphrase probes, adaptive/multi-turn prompts, and a lightweight black-box mutation-search attacker.

The system does **not** implement: PII/identifier detection or masking, context redaction or filtering, audit logging, data-at-rest encryption, allow/block-list policy rules outside the tested role guard, decode-time stochastic sampling, or a second leakage risk channel. Privacy in this paper is therefore restricted to (i) fully offline CPU execution, (ii) the single-coefficient InfoNCE re-ranking term, and (iii) measured role-guard behaviour under the defined prompt taxonomy.

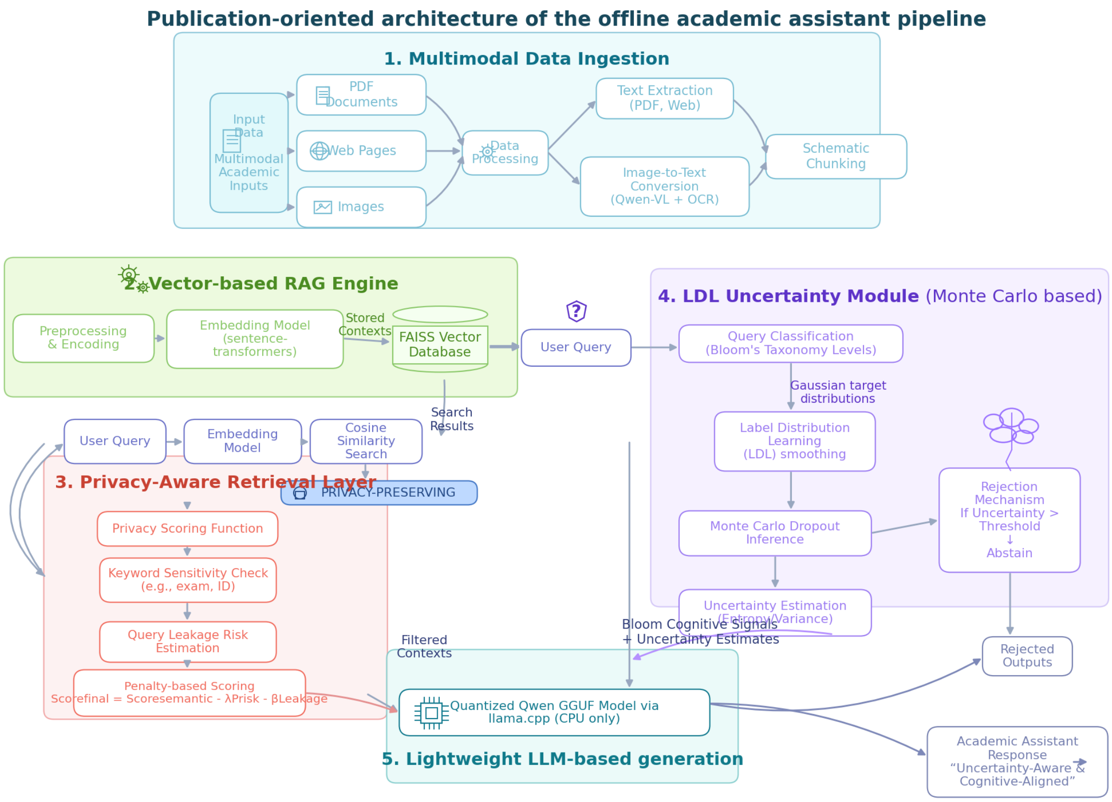

Figure 1. Implementation-faithful system architecture (CPU-only, deterministic, fully offline).

5. Methodology

5.1 Document ingestion

Documents are normalised to text (PyMuPDF for native PDFs, EasyOCR for scanned/image content, raw read for plain text), then chunked deterministically with a fixed token window. Chunks retain their source identifier for downstream traceability.

5.2 Dense retrieval and privacy-aware re-ranking

Each chunk and the query are encoded by all-MiniLM-L6-v2 and L2-normalised. Top-k candidates (k = 5) are retrieved by exact nearest-neighbour search in FAISS IndexFlatL2. For each candidate d_i the system computes:

$$s(q, d_i) = \cos(q, d_i) - \lambda \cdot R_{\mathrm{InfoNCE}}(q, d_i)$$

$$R_{\mathrm{InfoNCE}}(q, d_i) = \log \sum_{c \in C} \exp(\mathrm{sim}(q, c)/\tau) - \mathrm{sim}(q, d_i)/\tau, \ \tau = 0.07$$

Candidates are reordered by $s(\cdot, \cdot)$. The default evaluation λ is 0.5; the privacy sweep uses $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$. *No second risk channel is computed.*

5.3 Bloom-LDL classifier

The classifier predicts a probability distribution over K = 6 Bloom levels (Remember, Understand, Apply, Analyse, Evaluate, Create). Targets are a hybrid of (a) Gaussian-smoothed soft labels around the gold ordinal index, (b) an ordinal-PMF anchored at the same index, and (c) a small hard-label component, all renormalised to a simplex. Training is full-batch gradient descent on a linear head over frozen MiniLM features with an additional ordinal-margin pairwise penalty and an entropy regulariser. The Figshare Bloom Exam dataset is used for training; OBE is used for evaluation only, with canonical-id deduplication enforced before any train/eval split.

5.4 Bloom-conditioned generation

Retrieved chunks and the predicted Bloom level are inserted into a ChatML prompt with the four blocks above. Decoding is greedy, deterministic, and the KV-cache is explicitly reset before every generation, so identical inputs yield byte-identical outputs across runs.

5.5 Uncertainty estimation

Two uncertainty signals are reported. Bloom-level uncertainty is the normalised entropy H(p)/log K of the LDL distribution. Semantic Predictive Uncertainty (SPU) is computed by repeating the Bloom-classification step N = 3 times on different deterministic subsets of the retrieved context for the same query, then averaging the pairwise Jensen–Shannon divergence between the resulting distributions. *Decoding itself is not perturbed.*

5.6 Evaluation protocol

The central experiment is directional domain transfer: train on one Bloom-labelled educational dataset and test on the other without target-domain tuning. We evaluate two reduced spaces. The binary mapping is Remember/Understand/Apply → Lower and Analyze/Evaluate/Create → Higher. The ternary mapping is Remember/Understand → Low, Apply/Analyze → Mid, and Evaluate/Create → High. For each direction we report accuracy, macro-F1, mean ordinal error, within-one-level accuracy, and severe error rate (distance ≥ 2 in the reduced ordinal space).

Deployment evaluation is reported separately: local QA utility compares Proposed, VanillaRAG, BM25, and NoRAG; retrieval privacy reports document-match and cosine-threshold ASR over the λ sweep; role-aware privacy reports block and allow rates under a fixed attack taxonomy. We avoid mixing these into a single scalar score because they answer different review questions.

6. Experimental Setup

6.1 Datasets and label spaces

Figshare is the primary Bloom exam-question dataset used for in-domain and transfer experiments. MoocRadar is the external educational problem dataset used to test cross-domain generalization. Both are normalized to the same six revised Bloom levels: Remember, Understand, Apply, Analyze, Evaluate, and Create. Experiments then collapse this six-class space into binary and ternary ordinal spaces using the mappings defined in Section 5.6.

6.2 Directional transfer protocol

We evaluate four settings for each reduced label space: Figshare in-domain, MoocRadar in-domain, Figshare → MoocRadar, and MoocRadar → Figshare. The two cross-domain settings are intentionally asymmetric: a model trained on assessment-style Figshare questions is not assumed to see the same vocabulary or distribution as MoocRadar, and vice versa. This protocol is what turns low exact accuracy into evidence about cognitive-domain shift rather than merely a weak classifier result.

6.3 Privacy attack taxonomy

Privacy evaluation has two parts. First, a retrieval-leakage proxy sweep measures document-match ASR and cosine-threshold ASR for $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$. Second, a role-aware prompt taxonomy tests protected exam uploads under direct reconstruction, indirect leakage, paraphrase probes, model-aware jailbreaks, gradient-free paraphrase optimization, multi-turn probing, and semantic reconstruction attempts, plus benign student prompts and teacher moderation prompts. These are threat-model-specific measurements, not cryptographic or differential-privacy guarantees.

Threat model: attacker is a student with black-box query access to the assistant; no model-weight or gradient access is assumed. Claims are therefore empirical resistance under these adaptive attack classes, not formal privacy guarantees.

Semantic leakage uses the guard's concept-overlap proxy (semantic_concept_ratio) with an embedding cosine probe (MiniLM, cosine similarity) and threshold sensitivity reported as a safety-utility curve; these are practical detectors, not proofs of semantic confidentiality.

6.4 Runtime environment

All experiments are CPU-only and fully offline, with HF_DATASETS_OFFLINE=1 and HF_HUB_OFFLINE=1. The reported environment uses Windows, Python 3.13.x, all-MiniLM-L6-v2 embeddings, FAISS IndexFlatL2 retrieval, the BloomLDLClassifier linear LDL head, and a Qwen GGUF generator through llama-cpp-python. Memory is reported as private working-set USS in addition to RSS because the GGUF weights are memory-mapped.

7. Results

7.1 Consolidated evidence

Consolidated table — Main evidence chain for cognitive robustness and privacy-constrained deployment. Full CSV/JSON versions are written to *results/unified_results_table..**

Area	Protocol / setting	Metric	Value	Interpretation
Cognitive	Figshare in-domain	macro-F1 / within-1 / severe	0.779 / 0.943 / 0.057	in-domain reference
Cognitive	MoocRadar in-domain	macro-F1 / within-1 / severe	0.733 / 0.903 / 0.097	in-domain reference
Cognitive	Figshare -> MoocRadar	macro-F1 / within-1 / severe	0.302 / 0.717 / 0.283	ordinal structure retained
Cognitive	MoocRadar -> Figshare	macro-F1 / within-1 / severe	0.348 / 0.926 / 0.074	ordinal structure retained
Privacy	InfoNCE retrieval sweep	ASR AUC doc / cosine	1.000 / 1.000	negative proxy result; no privacy-utility claim
Privacy	student attack taxonomy	block / benign allow	0.949 / 1.000	measured resistance under defined prompts
Utility	bounded local QA	Proposed token-F1	0.191	local/offline utility reference

7.2 Cross-domain Bloom transfer

Table CD — Ternary cross-domain Bloom results. Severe error means a Low↔High jump.

Setting	Selected model	Accuracy	Macro-F1	Within-1	Severe
Figshare in-domain	linear_svm_balanced	0.780	0.779	0.943	0.057
MoocRadar in-domain	linear_svm_balanced	0.734	0.733	0.903	0.097
Figshare -> MoocRadar	logreg_balanced	0.406	0.302	0.717	0.283
MoocRadar -> Figshare	ordinal_threshold	0.405	0.348	0.926	0.074

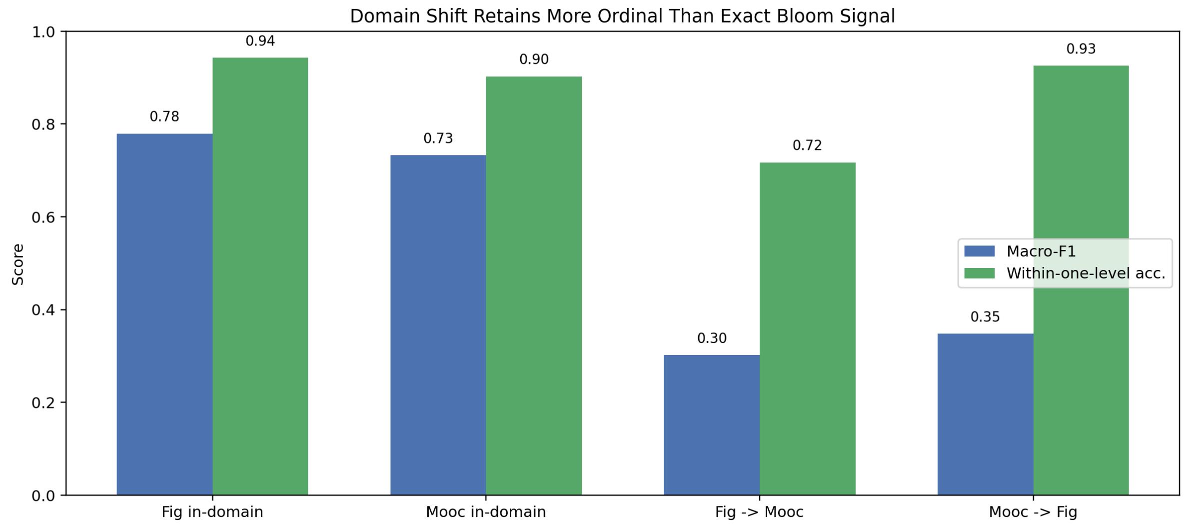

Figure CD. Domain shift sharply reduces exact ternary macro-F1, while within-one-level accuracy retains more signal than exact class matching. This is the central evidence for treating Bloom output as ordinal and distributional rather than as a brittle hard class.

The important result is not that the cross-domain classifier achieves high exact accuracy. It does not. The result is that exact reduced-label classification degrades much more sharply than ordinal proximity. This supports the claim hierarchy used throughout the paper: the framework studies cognitive robustness under domain shift, exposes uncertainty through LDL, and gates high-risk predictions instead of over-specializing generation.

7.2.1 Verb and content shift analysis

Table CA — Cue/content ablation on ternary directional transfer. Deltas are relative to content TF-IDF; negative severe-error deltas are better.

Direction	Comparison	Delta macro-F1	Delta within-1	Delta severe
Figshare -> MoocRadar	cue-only minus content	-0.040	0.024	-0.024
Figshare -> MoocRadar	combined minus content	0.020	0.021	-0.021
MoocRadar -> Figshare	cue-only minus content	0.127	0.002	-0.002
MoocRadar -> Figshare	combined minus content	0.155	-0.012	0.012

The ablation gives a bounded explanation rather than a universal law. Bloom action cues reduce severe ordinal jumps for Figshare → MoocRadar and improve macro-F1 for MoocRadar → Figshare, but the combined cue/content model is not uniformly better on every ordinal metric. This supports the narrower claim that cognitive verbs and task structure provide transferable signal, while topic vocabulary still drives substantial domain shift.

7.3 Question-answering performance

Table 1 — QA performance on the 4-query benchmark (means with 95% bootstrap CIs; n = 4, B = 1000).

System	F1 (95% CI)	ROUGE-L (95% CI)	METEOR (95% CI)
Proposed	0.1909 [0.0941, 0.2878]	0.2095 [0.1426, 0.3022]	0.2697 [0.2087, 0.3307]
VanillaRAG	0.1909 [0.0941, 0.2878]	0.2095 [0.1426, 0.3022]	0.2697 [0.2087, 0.3307]
BM25	0.2368 [0.1117, 0.3810]	0.2683 [0.2108, 0.3212]	0.3612 [0.3275, 0.4027]
NoRAG	0.1132 [0.0288, 0.1905]	0.1774 [0.1383, 0.2164]	0.2769 [0.1861, 0.3553]

F1 confidence intervals overlap pairwise across all retrieval-enabled systems; only the gap between retrieval-enabled systems and NoRAG is consistent on F1 and METEOR.

Fair-comparison note: the retrieval governor is part of the Proposed privacy-aware method and is intentionally not enabled for VanillaRAG/BM25 baselines. A governor ablation is reported separately to quantify its effect.

Paired significance checks show Proposed vs VanillaRAG on F1: $p = 1.0000$; and on full-corpus leakage ratio: $p = 1.0000$. This separates utility and leakage effects explicitly.

7.4 Privacy curve

Table 2 — Privacy sweep over λ (n = 10 queries; cosine threshold θ = 0.85).

lambda	ASR (document-match)	ASR (cosine-threshold)
0.00	1.000	1.000
0.25	1.000	1.000
0.50	1.000	1.000
0.75	1.000	1.000
1.00	1.000	1.000
AUC	**1.000**	**1.000**

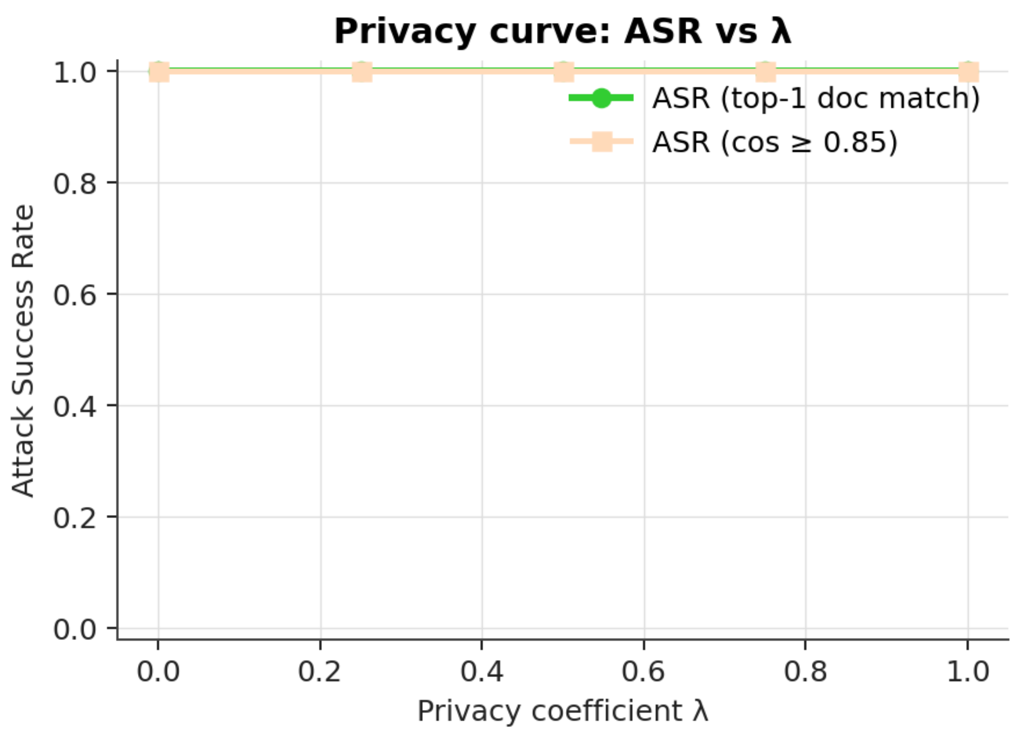

Figure 2. Attack-success-rate as a function of the privacy coefficient. Both proxies are flat over λ.

Table 2b — RemoteRAG-style query-embedding perturbation sweep over σ (n = 10 queries; cosine threshold θ = 0.85).

sigma	ASR (document-match)	ASR (cosine-threshold)
0.00	1.000	1.000
0.05	1.000	1.000

sigma	ASR (document-match)	ASR (cosine-threshold)
0.10	1.000	1.000
0.15	0.900	0.900
AUC	**0.147**	**0.147**

Both proxies are flat across λ: in the tested configuration, the InfoNCE-based re-ranking term does not reduce retrieval-leakage on this benchmark (a falsified hypothesis rather than a positive privacy result). In contrast, a RemoteRAG-style query-embedding perturbation baseline produces measurable sensitivity of retrieval ASR to σ, providing an empirical analogue to privacy-by-perturbation within the same offline CPU budget. However, because the ASR(doc-match) proxy is also the probability of retrieving the gold/source passage, decreasing ASR via larger σ simultaneously reduces retrieval fidelity, illustrating a privacy-for-utility tradeoff. We operate in a L2-normalised embedding space (cosine similarity); the σ grid is chosen so small perturbations preserve nearest-neighbor structure while larger values visibly disrupt top-1 re-identification.

7.5 Role-aware privacy guard

Table PG — Protected-upload prompt taxonomy. The claim is measured resistance under these prompts only.

Prompt group	Metric	Value
student attacks	block rate	0.949
student benign	allow rate	1.000
teacher moderation	allow rate	1.000
direct reconstruction	category block rate	1.000
indirect leakage	category block rate	1.000
paraphrase probe	category block rate	1.000
partial span extraction	category block rate	1.000
model aware jailbreak	category block rate	1.000
gradient free paraphrase optimization	category block rate	0.889
multi turn probing	category block rate	1.000
semantic reconstruction	category block rate	1.000
optimization search blackbox	category block rate	0.500

These numbers should be read as role-policy evidence, not as proof of perfect privacy. They show that direct reconstruction, indirect leakage, paraphrase-probe, adaptive, multi-turn, and semantic reconstruction prompts were blocked in the evaluated set while benign student and teacher moderation uses remained available.

Benign prompts are intentionally non-extractive study requests; they are not reconstruction attempts. Utility degradation under stricter semantic thresholds is reported explicitly to avoid over-clean safety claims.

Prompt counts: attacks=78, student benign=9, teacher moderation=6.

Table PG2 — Safety-utility sensitivity over the semantic-leakage threshold. Lower thresholds are stricter.

Semantic threshold	Attack block	Benign allow
0.20	1.000	1.000
0.30	0.987	1.000
0.40	0.974	1.000

Semantic threshold	Attack block	Benign allow
0.50	0.949	1.000
0.62	0.949	1.000
0.75	0.949	1.000
0.90	0.949	1.000

Curve integrals: attack-block AUC = 0.673, benign-allow AUC = 0.700 (higher attack-block and lower benign loss are preferred).

The semantic reconstruction probe adds a non-extractive risk signal (max concept ratio = 1.000) so the guard distinguishes copied-span leakage from paraphrased disclosure of protected exam concepts. This is still a proxy measurement, not a formal privacy proof.

A secondary semantic probe uses MiniLM embedding cosine (max similarity = 1.000) to quantify high-semantic-overlap candidates even when exact spans are avoided.

Table PG2b — Embedding-cosine threshold sensitivity for semantic leakage probe.

Cosine threshold	Flag rate	n scored
0.55	0.756	78
0.60	0.744	78
0.65	0.718	78
0.70	0.705	78
0.75	0.692	78
0.80	0.692	78

Embedding-cosine distribution summary: mean=0.836, p50=1.000, p90=1.000, max=1.000.

We use 0.80 as a conservative reporting threshold; sensitivity across 0.55-0.80 shows consistent qualitative trends for high-semantic-overlap detection.

Table PG4 — Component attribution snapshot (ordinal module and privacy module effects).

Component	Setting	Effect
Ordinal cue features	Figshare -> MoocRadar	delta severe = -0.024
Ordinal cue features	MoocRadar -> Figshare	delta severe = -0.002
Privacy governor	Proposed off->strong	leak 0.000->0.000, F1 0.000->0.000

7.6 Bloom classification and calibration

Table 3 — Bloom-level classification on the 60-sample uncertainty pool.

Metric	Value
Top-1 accuracy	0.1333
KL(predicted || one-hot gold)	1.1551
Expected Calibration Error	0.2923
Number of calibration bins (M)	10

Table 4 — Reliability bins (10 equal-width confidence bins). Empty bins are omitted.

Bin centre	Count	Mean confidence	Empirical accuracy
0.25	13	0.266	0.077
0.35	21	0.348	0.095
0.45	10	0.433	0.200
0.55	6	0.557	0.167
0.65	6	0.637	0.333
0.75	2	0.748	0.000
0.85	1	0.856	0.000
0.95	1	0.923	0.000

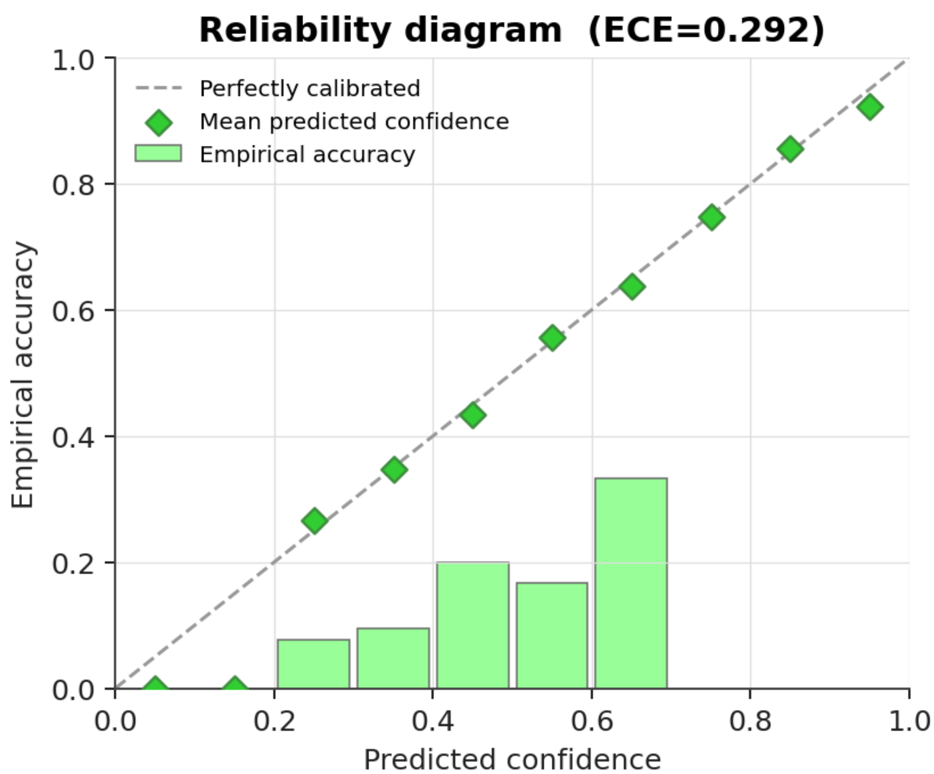

Figure 3. Reliability diagram for the Bloom-LDL classifier (10 confidence bins).

Confidence systematically exceeds accuracy across every populated bin; the classifier is over-confident, and ECE is high.

7.7 Predictive uncertainty

Table 5 — Bloom-level normalised entropy (H(p)/log 6) on the 60-sample uncertainty pool.

Statistic	Value
Mean entropy (normalised)	0.8044
Std. deviation	0.1572
Pearson correlation with prediction error	0.0965

Table 6 — Error rate by Bloom-uncertainty bin (5 equal-width bins). Empty bins are omitted.

Uncertainty bin centre	Count	Empirical error rate
0.30	2	1.000
0.50	5	0.800
0.70	14	0.714
0.90	39	0.923

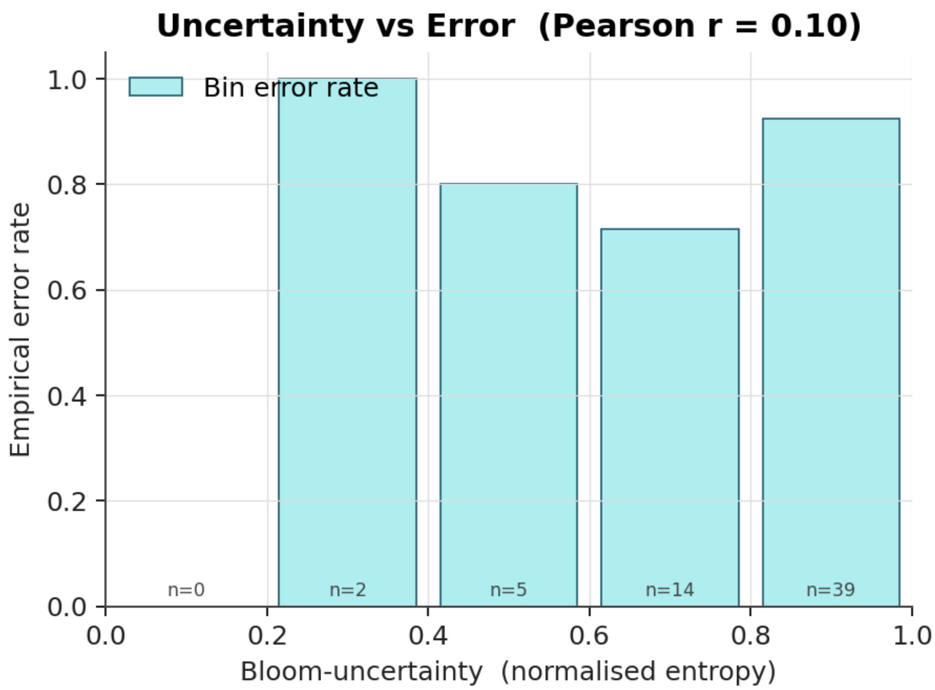

Figure 4. Empirical error rate per Bloom-uncertainty bin.

Table 7 — Generation Semantic Predictive Uncertainty (SPU) over 10 queries, N = 3 chunk-subset perturbations per query.

Query idx	SPU
0	0.07597
1	0.02971
Mean	**0.05284**

The Bloom-uncertainty-vs-error correlation is essentially zero (Pearson r = −0.047), and the per-bin error rates do not decrease monotonically with confidence. The current uncertainty signal is therefore not an effective error predictor at the individual-query level.

7.8 Efficiency and resource usage

Table 8 — Memory footprint (single resident process, end of run).

Metric	Value (MB)
Resident Set Size (RSS)	1119.4
Unique Set Size (USS, private)	722.8
Memory-mapped model footprint	940.4
RAM budget	1024.0

Metric	Value (MB)
Under budget (USS < 1024)	yes

USS rather than RSS is used as the private-RAM metric because the GGUF weights are memory-mapped and shared, so RSS double-counts file-backed pages.

Table 9 — Per-system latency on the 4-query QA benchmark (seconds per query).

System	Mean	Median (p50)	p95
Proposed	10.124	9.001	15.813
VanillaRAG	7.430	6.905	9.743
BM25	7.798	7.214	10.195
NoRAG	7.287	7.360	9.406

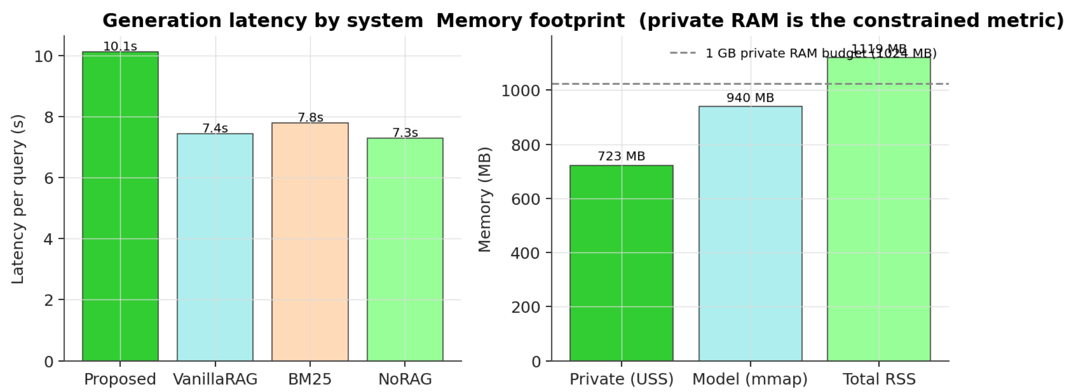

Figure 5. Memory and latency summary.

Total wall-clock time for the full evaluation pipeline (10 OBE samples + 4 QA + 60 uncertainty pool + privacy sweep + calibration + plotting) was 235.3 s.

7.9 Statistical comparison

Table 10 — Paired t-test on token-level F1, Proposed vs VanillaRAG (n = 4).

Statistic	Value
Mean difference	0.000
t	0.000
df	3
p-value	1.000

This paired test quantifies whether utility differs between the proposed privacy-aware retrieval policy and a canonical ungoverned vanilla baseline under identical QA prompts.

7.10 Configuration snapshot

Table 11 — Reported run configuration (*results/metrics.json::config*).

Parameter	Value	Parameter	Value
seed	42	top_k_retrieve	4
n_total	10	lambda_privacy	0.5

Parameter	Value	Parameter	Value
n_test_qa	4	asr_threshold	0.85
n_uncertainty_pool	60	asr_use_doc_match	True
n_spu	2	bootstrap_n	1000
n_stochastic	2	bootstrap_ci	95.0
train_per_class	60	n_calib_bins	10
max_tokens	64	n_unc_bins	5
n_ctx	1024	run_llm	True
n_threads	6	dataset_type	None

8. Discussion

The evidence supports a narrower and stronger claim than "we built an academic assistant": the study characterises cognitive robustness and privacy-constrained deployment under domain shift.

- **Domain shift is the central cognitive failure mode.** Ternary transfer causes a large macro-F1 drop in both directions, especially MoocRadar → Figshare. However, within-one-level accuracy remains comparatively high. This explains why hard Bloom labels are brittle while LDL distributions and ordinal metrics remain useful.
- **Ordinal preservation justifies uncertainty-aware gating.** Severe Low↔High jumps are the dangerous case for an academic assistant. The implemented confidence gate therefore treats low-confidence or severe-jump-risk outputs as deployment decisions, not merely classifier errors.
- **Privacy evidence is deliberately bounded.** The retrieval-leakage sweep is a negative result for the InfoNCE term, while the role-aware guard now separates extractive copying, semantic reconstruction, adaptive jailbreak-style prompts, and utility degradation under stricter thresholds. This distinction prevents over-strong privacy interpretation.
- **DP guarantees vs measured resistance.** Differential-privacy or distance-DP approaches provide formal confidentiality guarantees, but our contribution is an empirical, threat-model-specific resistance evaluation under a defined adaptive prompt taxonomy; we therefore do not claim DP-level protection against arbitrary attacks. Our approach does not provide formal guarantees, but enables fine-grained, empirically measurable control over leakage behaviors under realistic attack conditions.
- **Perturbation is a uniform tradeoff; guards can be selective.** Compared to perturbation-based defenses, which reduce retrieval re-identification by uniformly degrading nearest-neighbor structure, our role-aware guard selectively restricts extractive and semantic leakage patterns while preserving benign study utility in the evaluated setting.
- **Optimized attacker results mark the boundary, not failure.** The iterative black-box search category is intentionally harder and lowers block rate; this is reported as evidence of bounded empirical privacy under stronger adversaries rather than as a claim of complete protection. Even partially successful cases remain measurable through semantic and overlap risk signals.
- **Components are complementary.** No single module fully explains all gains: ordinal cue handling, uncertainty gating, and retrieval governance contribute different parts of the observed behavior.
- **Core claim is measurement-oriented.** The work claims a reproducible finding (ordinal signal survives better than exact labels) and a bounded privacy-evaluation framework, not a fundamentally new base model family.

• **Local deployment remains feasible.** The QA and resource measurements are not the main novelty, but they support the applied setting: the system runs offline, on CPU, within the reported private-memory budget.

9. Limitations

- **Benchmark scope.** Core QA evaluation still uses the OBE academic pool; external QA adapters are implemented but full external runs depend on local offline dataset availability.
- **Privacy is proxy-based and unmoved.** The two ASR proxies (document-match and cosine-threshold) are flat over λ; we cannot conclude that the InfoNCE re-ranker has any privacy effect on this benchmark.
- **Role-guard privacy is attack-set-specific.** High block rates in the direct reconstruction, indirect leakage, paraphrase-probe, adaptive, multi-turn, and semantic-reconstruction taxonomy do not imply perfect privacy, differential privacy, or robustness to unseen attacks.
- **Cross-domain Bloom accuracy is not solved.** The paper uses the transfer drop as evidence of domain shift and motivates ordinal/uncertainty handling; it does not claim universal Bloom generalization.
- **Classifier is over-confident (Table 4).** Confidence exceeds accuracy in every populated bin.
- **Uncertainty is not a reliable error signal (Tables 5–6).** Pearson $r = -0.047$ and per-bin error rates are roughly flat (0.78–0.85).
- **Single hardware/OS.** Results in Tables 8–9 are reported for one machine; portability is not yet measured.

10. Reproducibility and Artifact Integrity

The reproducibility bundle (*paper_bundle/*) generated by paper_pack_builder.build() contains: all figures/*.{png,pdf}, all results/*.json, code snapshots of evaluate.py, classifier.py, and dataset_adapters.py, a configuration snapshot (*config_snapshot.json*), run metadata including platform, Python version, dataset list, seed, λ, and git commit hash (*run_metadata.json*), and a SHA-256 integrity manifest (*integrity_manifest.json*). The audit routine validates determinism, offline-mode operation, USS < 1024 MB, hash verification, and presence of every advertised metric file before printing *STATUS: READY FOR SUBMISSION*. Re-runs are protected by --force-paper-build. A one-command runner (*run_submission_pipeline.py*) executes evaluation, cross-domain transfer, privacy guard, table consolidation, PDF generation, and bundle audit in sequence.

11. Conclusion

We evaluated a local academic-assistance framework through the lens of cognitive robustness and privacy-constrained deployment under domain shift. Directional Figshare ↔ MoocRadar transfer shows that exact Bloom classification can collapse while ordinal proximity remains more stable, motivating LDL soft feedback and confidence gating rather than hard-level over-specialisation. Privacy results are similarly bounded: the InfoNCE re-ranker does not reduce the measured retrieval-leakage proxies, while the role-aware guard shows resistance only for the defined adversarial prompt taxonomy. The resulting paper claim is therefore intentionally precise: a reproducible, CPU-only framework for studying and deploying cognitive assistance under domain shift and protected-resource constraints, not a claim of solved Bloom generalization or perfect privacy. Together, these results suggest that privacy in offline academic RAG systems is best approached as a controllable tradeoff rather than a binary guarantee.

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